

Math 445/545. Winter 2026. Practice problems for the Final.

The problems below illustrate various techniques you are supposed to know for the final. Some of the problems below are more difficult than the ones you will see on the final.

Problem 0. Please review Midterm 1, Midterm 2 (including practice problems).

Problem 1. Find the group $\mathbb{Z}^3/H$ where H is generated by $(6, 8, 12)$ and $(4, 10, 14)$.

Problem 2. Let A be a square integral $n \times n$ matrix and let $B = \mathbb{Z}^n/H$ where H is generated by the rows of A . Prove that B is finite if and only if $\det A \neq 0$. If B is finite prove that order of B is absolute value of $\det A$.

Problem 3. Prove that the group $\mathbb{Z} \times \mathbb{Z}/\langle(a, b)\rangle$ is cyclic if and only if a and b are relatively prime. How to find a generator of this group?

Problem 4. Let N be a subgroup of $\mathbb{Z}^{10}$ generated by 7 elements. Prove that $\mathbb{Z}^{10}/N$ is infinite.

Problem 5. (a) What is the number of ideals in the ring $\mathbb{R}[x]/(x^3 + x + 1)$?

(b) What is the number of ideals in the ring $\mathbb{Z}/1000\mathbb{Z}$?

Problem 6. What is the number of irreducible polynomials of the form $x^2 + ax + b$ in the ring $\mathbb{Z}_p[x]$? (Hint: First find the number of reducible polynomials of this form.)

Problem 7. Are the rings $\mathbb{Z}[x]/(x^2 - 1)$ and $\mathbb{Z} \oplus \mathbb{Z}$ isomorphic?

Problem 8. Let W be a 6-dimensional vector space.

a) Prove that for any 4-dimensional subspaces $W_1, W_2 \subset W$ the intersection $W_1 \cap W_2$ contains a non-zero vector.

b) Is it true that the intersection $W_1 \cap W_2 \cap W_3$ of any three 4-dimensional subspaces $W_1, W_2, W_3 \subset W$ contains a nonzero vector?

Problem 9. Let V be a n -dimensional vector space over the finite field $\mathbb{Z}_p$.

a) How many vectors are in V ?

b) How many 1-dimensional subspaces does V have?

Problem 10. How many elements are in $\mathbb{Z}[\sqrt{-2}]/(7 + 5\sqrt{-2})$?

Problem 11. Let $0 \neq a + bi \in \mathbb{Z}[i]$. Prove that the ring $\mathbb{Z}[i]/(a + bi)$ is finite and find its cardinality.

Problem 12. (a) Prove that $\mathbb{Z}[x]/(x^5 + 3x - 5)$ is finitely generated when considered as an abelian group.

(b) Prove that $\mathbb{Z}[x]/(2x^2 + 1)$ is not finitely generated when considered as an abelian group.